IE University
Master in Business Analytics & Data Science

Master's Thesis

Mean, Variance, and Market Efficiency in the Silver Market

Forecasting Weekly Returns and Realised Volatility with Classical, Machine-Learning, and Deep-Learning Models

Asier Ugarteche Perez

Supervisor: Prof. Dae-Jin Lee

Academic Year 2025/2026

Abstract

This thesis asks how much of silver's short-term price behaviour can be forecast, and frames the question through the efficient-market hypothesis (EMH). Silver is modelled at the weekly (Friday-to-Friday) horizon over 2015–2026, and the analysis is factorised into the two moments: the conditional *mean* of returns and the conditional *variance*—efficiency constrains the former but not the latter. For the mean, a deliberate ladder of models of increasing flexibility—ARIMA/ARIMAX, VAR, mixed-data-sampling (MIDAS) regressions, random forests, gradient boosting, and an LSTM—is tested against a common random-walk-with-drift benchmark, expanding the information set from silver's own past (weak-form EMH) to a broad set of public information (semi-strong-form EMH): cross-asset returns, silver futures positioning, macroeconomic releases, technical indicators, and sentiment from Reddit and financial news. For the variance, weekly realised volatility is modelled with GARCH and the heterogeneous autoregressive (HAR) model, and the question of interest is whether sentiment improves the forecast. The central finding is a single coherent statement: no model beats the drift benchmark for the conditional mean by a significant margin—weak- and semi-strong-form efficiency hold for weekly silver returns—yet the conditional variance is strongly and significantly forecastable, with sentiment adding a small but robust increment. Predictable volatility alongside an unpredictable mean is exactly what the EMH permits, so the two results are complementary rather than contradictory.

Contents

1 Introduction

Silver lives a double life. It is an industrial input—roughly half of annual demand goes into photovoltaics, electronics, brazing alloys, and medical uses, tying its price to the global manufacturing cycle—and, at the same time, a monetary metal held as a store of value, an inflation hedge, and a more volatile cousin of gold. On top of both sits a retail following that can sometimes turn silver into a meme trade: in late January 2021, Reddit's r/WallStreetSilver drove spot from roughly \$25 to \$30 an ounce within days before it faded. This blend of industrial demand, monetary hedging, and episodic retail attention makes silver an unusually rich laboratory for the oldest question in empirical finance: how much of a price can be forecast?

That question has a canonical answer, and it is mostly negative. The efficient-market hypothesis (Fama, 1970, 1991) holds that prices already embed available information, so successive price changes are approximately unpredictable. It is conventionally stated in three nested forms, graded by the information set a forecaster is allowed to use:

- **Weak form** — past prices and returns carry no exploitable signal.
- **Semi-strong form** — no *public* information does either: not prices, fundamentals, news, or macroeconomic data.
- **Strong form** — not even *private*, inside information does.

Each form nests the one before it, so the claims grow progressively stronger. For a liquid, globally traded metal, weak- and semi-strong-form efficiency are the natural hypotheses to take to the data, and they are the two this thesis tests.

Empirically, silver's *price* behaves like a random walk with drift: it is non-stationary, and shocks accumulate rather than mean-revert (Figure 1). Returns, however, are stationary—they fluctuate around a stable long-run mean rather than wandering without bound. The object worth modelling is therefore not the price level but its increment, the weekly log-return $r_t = \log P_t - \log P_{t-1}$; stationarity is what makes the forecasting problem well-posed, giving a fixed quantity to predict and a natural yardstick—that same prevailing average—to judge any forecast against.

The weekly (Friday-to-Friday) horizon is a deliberate choice of frequency, made where the information aligns. Futures-market positioning is reported weekly and macroeconomic releases monthly, while daily sentiment is too sparse and noisy to inform a forecast and daily returns are dominated by microstructure noise; a weekly horizon is coarse enough to average out that noise yet fine enough to leave roughly six hundred observations over 2015–2026—enough to train and properly out-of-sample-test the machine-learning models. Weekly realised volatility, the target of the second-moment analysis, is in turn aggregated from daily returns.

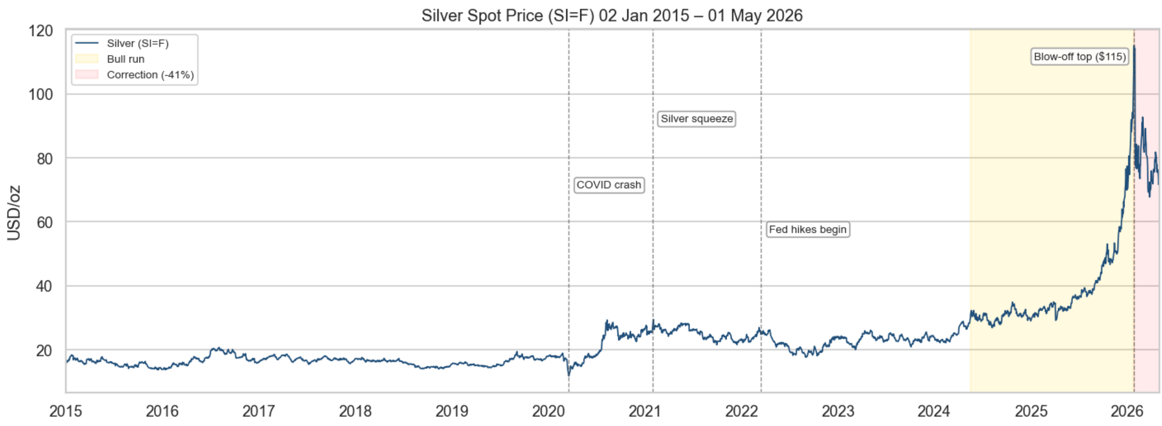

Figure 1: Silver front-month futures (SI=F), daily, January 2015–May 2026. The annotated episodes punctuate the sample: the COVID-19 crash (March 2020), the r/WallStreetSilver squeeze (January 2021), the start of the Federal Reserve's hiking cycle (March 2022), and the 2024–2026 bull run that carried spot to an all-time high near \$115/oz before a sharp (-41%) correction. The level is non-stationary and shocks accumulate; the thesis models its stationary weekly log-return rather than the level itself.

The efficiency question then becomes whether r_t can be forecast—and silver's own past offers little to go on. Weekly returns carry almost no linear autocorrelation, so a linear own-history model such as ARIMA has little to grip; a formal model search duly collapses it to a constant mean. Yet the returns are not independent noise either, because their *magnitude* is persistent—volatile weeks cluster together—so structure remains, only in the variance rather than the mean. The thesis follows this thread: it asks first whether silver's own history forecasts the return, then whether the broad public-information set a real forecaster would possess does—cross-asset returns (gold, the US dollar, copper, equities, oil, and the VIX), futures-market positioning, macroeconomic releases, technical indicators, and retail and news sentiment—and, separately, whether the predictable variance can be modelled.

Research questions. The thesis is built around four questions:

- **Q1 (weak form).** Are weekly silver returns forecastable from their own past?
- **Q2 (semi-strong form).** Does expanding the information set to public covariates help—and does it matter whether the relationship is assumed linear or allowed to be nonlinear?
- **Q3 (the second moment).** Is weekly realised volatility forecastable, and if so by how much?
- **Q4 (sentiment).** Does retail and news sentiment add incremental predictive

content to either the conditional mean or the conditional variance?

Approach. To answer these questions, the analysis works at the weekly (Friday-to-Friday) horizon and estimates a deliberate ladder of models of increasing flexibility against a common benchmark and a unified evaluation protocol. The benchmark throughout is the prevailing-mean "drift" forecast—the random-walk-with-drift, equivalent to ARIMA(0,0,0)—which is the correct null for a return target (Campbell & Thompson, 2008; Welch & Goyal, 2008). For the conditional mean, the ladder runs from classical time-series models (ARIMA/ARIMAX, VAR, and MIDAS regressions) through tree ensembles (random forests and gradient boosting) to a recurrent neural network (LSTM); for the conditional variance it runs GARCH and the heterogeneous autoregressive (HAR) model of Corsi (2009) alongside the same tree ensembles.

2 Conceptual Framework

This section fixes the notation used throughout and states, precisely, the nulls that each model class is built to test. The aim is to keep the language exact without overclaiming: the empirical results are framed as tests of the conditional mean and the conditional variance of weekly silver returns, not as claims about the full return distribution.

2.1 Prices, returns, and the random walk

Let P_t be the silver price and $p_t = \log P_t$ the log price. The weekly log-return is the increment in the log price,

$$r_{t+1} = p_{t+1} - p_t, \qquad \text{equivalently} \qquad p_{t+1} = p_t + r_{t+1}, \tag{1}$$

so the price level and the return are two views of the same process: the level accumulates shocks, while returns *are* the shocks. A random walk with drift writes the level as $p_{t+1} = p_t + \mu + \varepsilon_{t+1}$ with $\mathbb{E}[\varepsilon_{t+1}] = 0$. Silver log prices are best described as *random-walk-like*—non-stationary, with shocks that accumulate—which is the reason the modelling target throughout is the stationary increment r_t rather than the level. The safer wording avoids the *strict* random walk: the innovations are not independent and identically distributed, because their variance clusters (Section 2.4).

2.2 The conditional mean and the information set

Let F_t denote the information available at time t. For the weak-form test, this is silver's own past, $F_t = \{r_t, r_{t-1}, r_{t-2}, \dots\}$. Returns satisfy the *martingale difference* property with respect to F_t if their conditional mean is constant,

$$\mathbb{E}[r_{t+1} \mid F_t] = \mu, \tag{2}$$

or, equivalently, the demeaned return $r_{t+1} - \mu$ is a *martingale difference sequence* (MDS),

$$\mathbb{E}[r_{t+1} - \mu \mid F_t] = 0. \tag{3}$$

Past returns do not move the expected next-week return away from its long-run average. This is a statement about the first moment *only*; it does not assert that returns are i.i.d. The semi-strong test expands the information set to public covariates,

$$G_t = \{F_t,\ \text{gold},\ \text{USD},\ \text{copper},\ \text{VIX},\ \text{oil},\ \text{COT},\ \text{macro},\ \text{technicals},\ \text{sentiment}\}, \tag{4}$$

and asks whether the martingale-difference property survives the larger conditioning set,

$$\mathbb{E}[r_{t+1} \mid G_t] = \mu. \tag{5}$$

ARIMA tests whether silver returns are a martingale difference with respect to their own past, Equation (2); ARIMAX and the other models test whether that property survives once the information set is expanded to public covariates, Equation (5). Operationally, these nulls are tested as "does the model beat the prevailing-mean (drift) benchmark out of sample?"—a forecast that beats the drift would contradict the corresponding null.

2.3 Linear and nonlinear alternatives

The alternative to Equation (5) can be specified narrowly or broadly. A linear model entertains an additive alternative,

$$\mathbb{E}[r_{t+1} \mid G_t] = \mu + \beta' G_t, \tag{6}$$

so ARIMAX, VAR, and MIDAS ask whether public covariates enter the conditional mean linearly and additively. Nonlinear models entertain the broader alternative

$$\mathbb{E}[r_{t+1} \mid G_t] = f(G_t), \tag{7}$$

in which the conditional mean can respond to thresholds and interactions among the covariates rather than to each one additively—a signal might be invisible in any single predictor yet surface in a combination of them. Random forests, gradient boosting, and the LSTM search this larger space. The logic of the ladder is therefore cumulative: if even flexible nonlinear models fail to beat the drift benchmark out of sample, the martingale-difference interpretation is *strengthened* rather than merely assumed from linear tests. A complementary own-history check, the generalised spectral test (Hong & Lee, 2003), probes for nonlinear departures from Equation (2) that linear autocorrelation diagnostics cannot see.

2.4 The conditional variance

Efficiency constrains Equation (5); it places no restriction on the conditional variance $\mathrm{Var}(r_{t+1} \mid F_t)$. The two can diverge: the mean may be (approximately) constant while the variance moves through time, which is exactly what volatility clustering means—a large move today says little about the sign or size of next week's return but does signal that next week is likely to be more volatile. Strict white noise would require independence in r_t, r_t^2, and $|r_t|$ alike; the MDS property in Equation (2) restricts only the first of these. Weekly silver returns are close to white noise in the mean but not strict white noise, because squared returns are strongly autocorrelated. The predictable structure is therefore concentrated in the second moment, and the squared return is its natural proxy: by definition

$$\mathbb{E}[r_t^2] = \mathrm{Var}(r_t) + \Big(\mathbb{E}[r_t]\Big)^2, \tag{8}$$

and at the weekly horizon the squared mean is negligible. For silver $\Big(\mathbb{E}[r_t]\Big)^2 \approx 7 \times 10^{-6}$ is only 0.4% of the variance $\mathrm{Var}(r_t) \approx 1.9 \times 10^{-3}$, so $\mathbb{E}[r_t^2] \approx \mathrm{Var}(r_t)$: at this frequency a squared return measures variance almost exactly. The volatility chapter (Section 7) builds on this, targeting weekly realised volatility aggregated from daily squared returns. Figure 2 shows both moments at once for the weekly silver return.

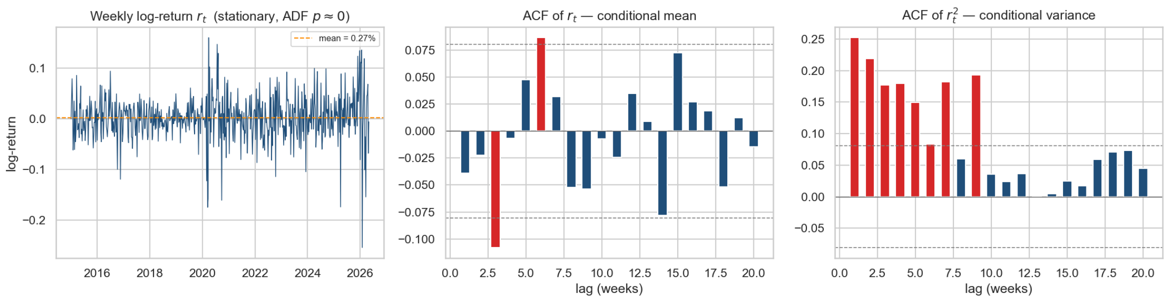

Figure 2: The two moments of the weekly silver log-return, 2015–2026. *Left:* the return series—stationary, fluctuating around a small positive drift (mean $\approx 0.27\%$ per week). *Centre:* the sample autocorrelation of returns lies almost entirely within the 95% white-noise band (dashed), with only isolated lags breaching it—consistent with the martingale-difference null of Equation (2), so a linear own-history model has little to exploit. *Right:* the autocorrelation of *squared* returns is large and positive across the first several lags—the volatility clustering of Section 2.4. The mean is close to unpredictable; the variance is not.

2.5 Direction versus magnitude

A final distinction governs how forecasts are scored. Directional accuracy asks whether $\text{sign}(\hat{r}_{t+1}) = \text{sign}(r_{t+1})$ more often than chance, which is *not* the same as beating the drift benchmark under squared-error loss: a model can call the sign correctly yet fail to improve out-of-sample R^2. Because efficiency is a conditional-mean claim, the magnitude axis (RMSE, out-of-sample R^2, and Diebold–Mariano against the drift) is load-bearing, while directional accuracy—and its significance test, Pesaran–Timmermann (Pesaran & Timmermann, 1992)—is a secondary, economically motivated lens. Directional gains are interesting but do not by themselves reject the martingale-difference null.

3 Literature Review

The thesis sits at the intersection of five strands of literature: the long empirical record on commodity-return predictability and market efficiency, mixed-frequency regression, machine learning for financial forecasting, realised-volatility modelling, and text-based sentiment. This section reviews each in turn, with an eye on the specific design choices they motivate in the chapters that follow.

3.1 Commodity-return predictability and market efficiency

The efficiency framing of Section 2 descends directly from Samuelson (1965) and the weak/semi-strong/strong taxonomy of Fama (1970), whose later survey (Fama, 1991) reframed the weak form more broadly as "tests for return predictability"—precisely the form the present exercise takes. For commodities specifically, the canonical stylised facts are discouraging for forecasters: Deaton and Laroque (1992) document spot prices that are extremely volatile, dominated by long quiet spells punctuated by sharp spikes, and difficult to distinguish from highly persistent, near-unit-root processes—which is why the stationary increment (the return), not the level, is the only defensible modelling target. At the same time, commodities do not move in isolation: Zhang and Wei (2010) find cointegration and price discovery between crude oil and gold, and Batten et al. (2010) show that the macroeconomic determinants of volatility differ across the precious metals, with silver responding to a different mix of monetary variables than gold. Together these findings motivate the cross-asset information set used here—gold, the dollar, copper, equities, the VIX, and oil—while cautioning against treating silver as a small gold.

Whether any of that public information translates into out-of-sample forecasting power is the subject of a debate the thesis borrows its evaluation protocol from. Welch and Goyal (2008) re-examine the full catalogue of variables claimed to predict the equity premium and find that essentially none beats the prevailing historical mean once evaluated out of sample; Campbell and Thompson (2008) reply that modest gains are possible, but only with theory-imposed restrictions, and propose the out-of-sample R^2 against the prevailing-mean forecast as the natural yardstick. That benchmark—the random-walk-with-drift, estimated as an expanding historical mean—and that metric are adopted unchanged in Section 5. Finally, Lo (2004) offers the adaptive-markets reading in which efficiency is not a binary property but an evolutionary, time-varying one, so that pockets of predictability can appear and decay; this is the natural frame for the episodic, non-replicating directional signals encountered in Section 6.

3.2 Mixed-frequency models (MIDAS)

Forecasting a weekly return with monthly macroeconomic releases and daily market data poses an alignment problem: aggregating the high-frequency series throws information away, while including every high-frequency lag as a free regressor invites massive parameter proliferation. Mixed-data-sampling (MIDAS) regression (Ghysels et al., 2004, 2007) resolves the trade-off by letting the high-frequency lags enter through a low-dimensional weight polynomial—typically the Beta or exponential-Almon family—so that a long lag window costs two or three parameters instead of dozens. Andreou et al. (2013) demonstrate the payoff in the direction most studied, using daily financial data to forecast quarterly macro aggregates; the present application inverts that direction, bringing monthly macro releases and daily cross-asset returns to bear on a weekly financial target. The `midasr` package (Ghysels et al., 2016) documents the now-standard estimation approach—nonlinear least squares over the weight parameters with the linear coefficients profiled out—which the Python implementation in this thesis follows.

3.3 Machine learning and deep learning for financial time series

Tree ensembles and recurrent neural networks instantiate the nonlinear alternative $f(G_t)$ of Section 2.3: random forests and gradient boosting capture thresholds and interactions without specifying them in advance, and the long short-term memory network (Hochreiter & Schmidhuber, 1997) adds explicit sequence memory, extracting its own representation of the recent past rather than consuming hand-picked lags. The applied literature is large and, on its face, encouraging—the systematic review of Sezer et al. (2020) catalogues hundreds of deep-learning forecasting studies reporting gains over classical models. It is also methodologically uneven: benchmarks are often weak or absent, evaluation windows short, and statistical significance rarely assessed, so reported out-performance frequently dissolves against an honest no-change or prevailing-mean benchmark. The design response in this thesis is to keep the model ladder flexible but the referee strict: every learner, however expressive, is scored against the same drift benchmark with the same Diebold–Mariano test, so that flexibility can only ever strengthen—never substitute for—the efficiency verdict.

3.4 Realised volatility and the HAR model

That the conditional variance, unlike the conditional mean, is forecastable has been textbook knowledge since Engle (1982) introduced ARCH and Bollerslev (1986) generalised it: volatility clusters, so its own past predicts it. The realised-volatility literature sharpened the target, replacing latent variance with a model-free estimate built from higher-frequency squared returns—here, weekly realised volatility from daily returns. Its workhorse forecaster is the heterogeneous autoregressive (HAR) model of Corsi (2009), which regresses next period's realised volatility on its own past averaged over short, medium, and long horizons; despite being estimable by OLS, the cascade of horizons approximates the long memory of volatility well enough that HAR remains the standard benchmark. Evaluation has its own subtlety: realised volatility is itself a noisy proxy of the latent target, and Patton (2011) shows that only a restricted family of loss functions—in practice MSE and QLIKE—ranks forecasts consistently under proxy noise, with QLIKE the more powerful in heavy-tailed samples. The volatility chapter therefore reports QLIKE-based Diebold–Mariano tests as primary. None of this strains the efficiency hypothesis, which constrains the mean and not the variance (Section 2.4): predictable volatility is permitted, and indeed expected.

3.5 Financial sentiment and investor attention

That media tone carries information for markets predates modern language models: Tetlock (2007), using a simple dictionary count of a Wall Street Journal column, found that media pessimism predicts downward pressure on prices followed by reversal. What has changed is measurement. Transformer language models (Devlin et al., 2019; Liu et al., 2019) replaced dictionaries with contextual representations, and domain-adapted variants—FinBERT for financial news (Araci, 2019) and Twitter-RoBERTa for social-media text (Loureiro et al., 2022)—now score sentiment with near-annotator accuracy on their home domains; both are used here, each on the text type it was trained for. On where such signals should help, the evidence points to the second moment: Audrino et al. (2020) find that sentiment and attention measures add significant predictive power for stock-market volatility—with attention the stronger of the two—but only marginal power for returns, and Smales (2021) documents investor attention, proxied by Google searches, as a state variable for market reactions under stress. Silver brings a natural laboratory for these attention effects: the January 2021 r/WallStreetSilver episode (Section 1) was an explicitly coordinated retail-attention shock, and the Reddit and news series collected here cover it. The volatility chapter's sentiment ablation is designed exactly along this literature's fault line: attention versus tone, variance versus mean.

4 Data

All models are estimated on a single weekly dataset spanning January 2015 to May 2026. The raw inputs come from seven source families—market prices, daily financial indicators, monthly macroeconomic releases, futures-market positioning, Reddit, financial news, and Google Trends—each collected at its native frequency and then consolidated onto a common Friday-to-Friday (W-FRI) calendar in one shared feature frame (Section 4.4). This section describes each source and the features derived from it; the model inputs are summarised in Table 2.

4.1 Price and cross-asset data

Daily closing prices for silver and six related assets—gold, the US dollar index, copper, the S&P 500, the VIX, and crude oil—are obtained from Yahoo Finance over 2015–2026 (2,854 trading days). Silver is the front-month futures contract, and its weekly log-return $r_t = \log P_t - \log P_{t-1}$, taken Friday-to-Friday, is the forecast target throughout. The six cross-asset series enter as their own weekly log-returns—the EXOG set—capturing silver's documented co-movement with the precious metals, the dollar, industrial demand, equities, and risk appetite (Batten et al., 2010; Zhang & Wei, 2010). Silver's own three weekly return lags (the AR block) and the gold/silver price ratio, z-scored on the training period (`gs_ratio_z`), complete the price-based features.

4.2 Macroeconomic and positioning data

Three daily financial indicators are taken from the Federal Reserve's FRED database: the ten-year TIPS real yield, the ten-year breakeven inflation rate, and initial jobless claims, each entered as its weekly change. Four monthly macroeconomic series—headline CPI, the federal funds rate, industrial production, and M2—are likewise from FRED. Because a month's figure is not public until weeks after its reference period, the macro series are not simply shifted but entered under a point-in-time availability rule: a monthly value is visible in week d only once it would actually have been released, and three publication lags of each series are retained ($*$`_mlag1/2/3`). Futures-market positioning comes from the CFTC Commitments of Traders report, published weekly; we use the managed-money and commercial net positions, each expressed as a share of open interest (`cot_mm_net_pct`, `cot_comm_net_pct`).

4.3 Sentiment data

Two text sources are scored for sentiment. Reddit posts from r/WallStreetSilver and r/Silverbugs, recovered back to 2015 through the Arctic Shift archive, are scored with a RoBERTa model fine-tuned on social-media text (Loureiro et al., 2022); financial-news headlines from GDELT are scored with FinBERT, a BERT model adapted to the financial domain (Araci, 2019). Each document's score is averaged to a daily index and then to a weekly mean (`reddit_sentiment`, `news_sentiment`). News coverage is the thinner of the two: GDELT begins only in late 2017 and has zero-article weeks even inside the test window, which is why the volatility chapter's sentiment study (Section 7) relies on Reddit attention and treats the news series as a documented data limitation. A separate Google Trends "silver" search-interest series is also collected as a broad retail-attention proxy for that chapter.

4.4 Feature frame and train/validation/test split

All sources are consolidated into one shared weekly frame of 591 W-FRI weeks. Aggregation from daily to weekly follows fixed rules: returns and FRED changes are summed within the week, sentiment is averaged, and levels (`gs_ratio_z`) and COT positioning take the week's last value. Every feature enters the frame raw and contemporaneous, with two exceptions: the macro block, which is publication-lagged as above, and a set of filter-selected best-lag columns used only for one ablation. Each model then applies its own one-week lag, so a feature observed at the Friday-$t-1$ close is used to forecast the return realised at the Friday-t close—there is no intra-week look-ahead.

The data are split chronologically, without shuffling and with boundaries on Fridays: 2015–2021 for training, 2022 for validation (model and hyperparameter selection), and 2023 to May 2026 as the held-out test window (Table 1). The split deliberately places the unusual 2024–2026 bull run in the test set, so that out-of-sample performance is judged across a genuine regime shift rather than an in-sample echo of it.

Table 1: Chronological train/validation/test split (W-FRI weeks).

Split	Window	Weeks	Role
Train	2015-01 – 2021-12	365	model fitting
Validation	2022	52	model & hyperparameter selection
Test	2023-01 – 2026-05	174	held-out evaluation
Total	2015 – 2026	591	

Table 2: Variables entering the analysis, by feature group. All features are observed at the Friday-$t-1$ close and lagged one week by each model before use, except the macro block (publication-lagged) and a best-lag ablation set. “EXOG” denotes the six cross-asset returns.

Group	Variables	Source	Freq.	Entered as
Target	`silver_return`	Yahoo Finance	Daily	weekly W-FRI log-return
AR	`silver_lag1/2/3`	Yahoo Finance	Daily	weekly log-returns, 1–3 wk lags
Cross-asset (EXOG)	`gold_return`, `copper_return`, `usd_return`, `sp500_return`, `vix_return`, `oil_return`	Yahoo Finance	Daily	weekly log-returns
Ratio (GS)	`gs_ratio_z`	Yahoo Finance	Daily	gold/silver ratio, z-scored on train
FRED daily	`real_rates_chg`, `breakeven_chg`, `jobless_chg`	FRED	Daily/Wk	weekly change
Positioning (COT)	`cot_mm_net_pct`, `cot_comm_net_pct`	CFTC	Weekly	net position / open interest
Sentiment	`reddit_sentiment`, `news_sentiment`	Reddit, GDELT	Daily	weekly mean of RoBERTa / FinBERT scores
Technical (TECH)	`macd_line`, `macd_hist`, `rsi_14w`, `mom_5w`, `roc_13/26/52w`, `price_ma13w`, `donchian_52w`, `bb_pct_b`	Yahoo Finance	Weekly	directional momentum / trend indicators
Macro (MACRO)	`cpi`, `fed_funds`, `ind_prod`, `m2`	FRED	Monthly	3 publication-availability lags each

5 Methodology

5.1 Forecasting target and benchmarks

The target for the conditional-mean analysis is the weekly silver log-return r_t of Section 2.1, forecast one week ahead. Two benchmarks frame every model. The first is a *naïve* forecast, $\hat{r}_{t+1} = r_t$, the persistence of last week's return; it is a weak reference for a near-white-noise target and is reported only for context. The load-bearing benchmark is the *drift*, the prevailing (expanding) mean of past returns,

$$\hat{r}_{t+1}^{\text{drift}} = \frac{1}{t}\sum_{s=1}^{t} r_s, \tag{9}$$

which is the random-walk-with-drift forecast and is identical to ARIMA$(0,0,0)$ on an expanding window. It is the correct null for a return target (Campbell & Thompson, 2008; Welch & Goyal, 2008): a model that cannot beat the drift out of sample carries no exploitable conditional-mean information, so "beating the drift" is the operational form of rejecting the efficiency nulls of Equations (2) and (5).

Models are evaluated by *walk-forward* one-step-ahead forecasting on the held-out 2023–2026 window (Section 4.4): at each week a model is fit on data through t and predicts $t+1$, so no future information enters a forecast. Where the model permits, two windowing schemes are compared—an *expanding* window that grows with the sample, and a *rolling* 100-week window that discards older data to track regime change. Re-estimation cadence is model-specific and noted below.

5.2 Conditional-mean models

The models form a deliberate ladder of increasing flexibility—from linear own-history, through linear public-information, to nonlinear machine learning—every rung judged against the same drift floor. Each exogenous predictor enters lagged one week (Section 4.4), and the ablation *ladder* adds feature groups cumulatively to a six-asset cross-return base (EXOG): the gold/silver ratio, technical indicators, monthly macro, daily FRED indicators, COT positioning, and Reddit and news sentiment, up to an all-features rung.

ARIMA / ARIMAX. An ARIMA(p, d, q) model with order chosen by AIC on the training and validation data; the search selects ARIMA$(0,0,0)$, so the best own-history model coincides with the drift and gives the cleanest weak-form test. ARIMAX adds the lagged exogenous groups as regressors and supplies the linear semi-strong test.

VAR. A vector autoregression of silver and the cross-asset returns with lag order by AIC, capturing linear feedback among the assets; Granger-causality and impulse-response diagnostics are reported descriptively.

MIDAS. Mixed-data-sampling regressions that bring higher-frequency predictors to the weekly target through a low-dimensional weight polynomial—the Beta and exponential-Almon families (Ghysels et al., 2004, 2007)—estimated by nonlinear least squares, with the weight family selected on the validation window. Two variants are run: monthly macro at three lags, and a multi-frequency model adding the six cross-asset returns at twenty daily lags (Andreou et al., 2013).

Random forest and gradient boosting. Nonlinear ensembles that admit thresholds and interactions among the lagged features—a random forest (Breiman, 2001) and gradient-boosted trees, XGBoost (Chen & Guestrin, 2016). Hyperparameters are grid-searched per ablation rung by blocked time-series cross-validation on train-plus-validation, and the model is re-fit every four weeks in the walk-forward.

LSTM. A long short-term memory recurrent network (Hochreiter & Schmidhuber, 1997) that reads a fixed window of recent weeks and learns its own representation of the past; a single recurrent layer with dropout and early stopping is used, given the small sample (a few hundred training sequences). A weekly-input model and a daily-input variant (a sixty-trading-day window) are estimated, the latter testing whether daily-resolution information helps forecast the weekly return.

The classical and tree models are run under both windowing schemes; the LSTM uses a single expanding-window training pass, since a 100-week rolling window leaves too few sequences to train stably.

5.3 Conditional-variance models

The second-moment analysis replaces the return with weekly *realised volatility.* Realised variance is additive across days, so the weekly figure is the square root of the summed daily squared returns,

$$\mathrm{RV}_t = \sqrt{\textstyle\sum_{i \in \text{week } t} r_i^2}, \tag{10}$$

a model-free volatility proxy. Its predictors are the heterogeneous-autoregressive (HAR) lags—one-week, monthly (four-week mean), and quarterly (twelve-week mean) trailing realised volatility—optionally extended with the six cross-asset realised volatilities, Reddit attention and sentiment-intensity measures, and Google Trends search interest; the last three form the volatility chapter's sentiment ablation, and Google Trends enters here only, not in the conditional-mean models.

GARCH. A GARCH(1,1) model (Bollerslev, 1986; Engle, 1982), fit by maximum likelihood and re-estimated walk-forward—the parametric benchmark for conditional variance.

HAR-RV and HAR-X. The heterogeneous autoregressive model of Corsi (2009), an OLS regression of next-week realised volatility on its three trailing averages; HAR-X adds the cross-asset, attention, and search predictors as ablation rungs.

Tree ensembles. Random forest and XGBoost on the HAR-plus-cross-asset feature set, isolating any nonlinear gain over the linear HAR-X.

5.4 Evaluation protocol and significance tests

All models are scored on the same 2023–2026 test window. Point-forecast accuracy is measured by RMSE and MAE—the primary metrics, since efficiency is a conditional-mean (magnitude) claim—complemented by directional accuracy (DA) and its magnitude-weighted form (WDA) for the returns, and by direction-of-change accuracy (DCA) for volatility. Effect size relative to the drift is summarised by the Campbell–Thompson out-of-sample R^2 (Campbell & Thompson, 2008).

Significance rests on three tests. The **Diebold–Mariano** test of equal predictive accuracy (Diebold & Mariano, 1995), with Newey–West variance, compares each model to the drift floor; for returns it is reported under squared-error loss (the headline) and absolute-error loss (a heavy-tail robustness check), and for volatility under the proxy-robust QLIKE loss (Patton, 2011), which squared-error DM is too low-powered to replace on heavy-tailed realised volatility. The **Pesaran–Timmermann** test (Pesaran & Timmermann, 1992) is the matching significance test for the directional (DA/WDA) reading. Finally, a generalised spectral test (Hong & Lee, 2003) checks the own-history return series for nonlinear departures from the martingale-difference null that linear autocorrelation cannot detect.

Two robustness layers accompany the headline tests: a breakdown by calendar sub-period across 2023–2026, to confirm that performance is not driven by a single regime, and a re-run of the full battery with the exceptional 2025 bull run excluded. The load-bearing verdict throughout is the Diebold–Mariano test against the drift floor; the directional results are a secondary, economically motivated lens.

6 The Conditional Mean: Forecasting Weekly Returns

6.1 Weak form: own-history models

6.2 Semi-strong form: public information

6.3 Direction versus magnitude

6.4 Robustness

7 The Conditional Variance: Forecasting Weekly Volatility

7.1 Realised volatility is forecastable

7.2 Cross-asset spillover and nonlinearity

7.3 Sentiment and attention

8 Discussion

9 Conclusion

Weekly silver returns are indistinguishable from a martingale difference under magnitude loss, satisfying weak- and semi-strong-form efficiency in the economically relevant sense; the residual directional dependence is statistically detectable but unexploitable, which efficiency theory permits; and its episodic, regime-bound character is consistent with—though not proof of—the adaptive-markets view (Lo, 2004) in which such efficiency is the equilibrium outcome rather than the assumption. One story, three layers, no tension.

Contributions. The thesis makes four contributions. First, it evaluates a full model ladder—classical, machine-learning, and deep-learning—for *both* moments of weekly silver returns under one drift benchmark and one significance battery, so that model families are compared on a level footing rather than across incommensurable setups. Second, it makes the conditional-mean / conditional-variance factorisation explicit, turning what could read as "returns failed, volatility worked" into a single coherent efficiency statement. Third, it builds a domain-appropriate sentiment pipeline—Twitter-RoBERTa for Reddit, FinBERT for financial news—and treats sentiment as a falsifiable ablation on both moments rather than assuming it helps. Fourth, it documents a carefully qualified set of negative results on semi-strong predictability of the mean, separating magnitude from direction and showing where an apparent directional signal fails to replicate across model classes.

References

Andreou, E., Ghysels, E., & Kourtellos, A. (2013). Should macroeconomic forecasters use daily financial data and how? *Journal of Business & Economic Statistics*, *31*(2), 240–251.

Araci, D. (2019). FinBERT: Financial sentiment analysis with pre-trained language models. *arXiv preprint arXiv:1908.10063*.

Audrino, F., Sigrist, F., & Ballinari, D. (2020). The impact of sentiment and attention measures on stock market volatility. *International Journal of Forecasting*, *36*(2), 334–357.

Batten, J. A., Ciner, C., & Lucey, B. M. (2010). The macroeconomic determinants of volatility in precious metals markets. *Resources Policy*, *35*(2), 65–71.

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, *31*(3), 307–327.

Breiman, L. (2001). Random forests. *Machine Learning*, *45*(1), 5–32.

Campbell, J. Y., & Thompson, S. B. (2008). Predicting excess stock returns out of sample: Can anything beat the historical average? *Review of Financial Studies*, *21*(4), 1509–1531.

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794.

Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, *7*(2), 174–196.

Deaton, A., & Laroque, G. (1992). On the behaviour of commodity prices. *Review of Economic Studies*, *59*(1), 1–23.

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of NAACL-HLT*, 4171–4186.

Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, *13*(3), 253–263.

Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of united kingdom inflation. *Econometrica*, *50*(4), 987–1007.

Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. *Journal of Finance*, *25*(2), 383–417.

Fama, E. F. (1991). Efficient capital markets: II. *Journal of Finance*, *46*(5), 1575–1617.

Ghysels, E., Kvedaras, V., & Zemlys, V. (2016). Mixed frequency data sampling regression models: The R package midasr. *Journal of Statistical Software*, *72*(4), 1–35.

Ghysels, E., Santa-Clara, P., & Valkanov, R. (2004). The MIDAS touch: Mixed data sampling regression models. *CIRANO Working Papers*.

Ghysels, E., Sinko, A., & Valkanov, R. (2007). MIDAS regressions: Further results and new directions. *Econometric Reviews*, *26*(1), 53–90.

Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, *9*(8), 1735–1780.

Hong, Y., & Lee, T.-H. (2003). Inference on predictability of foreign exchange rates via generalized spectral and nonlinear time series models. *Review of Economics and Statistics*, *85*(4), 1048–1062.

Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., & Stoyanov, V. (2019). RoBERTa: A robustly optimized BERT pretraining approach. *arXiv preprint arXiv:1907.11692*.

Lo, A. W. (2004). The adaptive markets hypothesis: Market efficiency from an evolutionary perspective. *Journal of Portfolio Management*, *30*(5), 15–29.

Loureiro, D., Barbieri, F., Neves, L., Anke, L. E., & Camacho-Collados, J. (2022). TimeLMs: Diachronic language models from Twitter. *arXiv preprint arXiv:2202.03829*.

Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, *160*(1), 246–256.

Pesaran, M. H., & Timmermann, A. (1992). A simple nonparametric test of predictive performance. *Journal of Business & Economic Statistics*, *10*(4), 461–465.

Samuelson, P. A. (1965). Proof that properly anticipated prices fluctuate randomly. *Industrial Management Review*, *6*(2), 41–49.

Sezer, O. B., Gudelek, M. U., & Ozbayoglu, A. M. (2020). Financial time series forecasting with deep learning: A systematic literature review 2005–2019. *Applied Soft Computing*, *90*, 106–181.

Smales, L. A. (2021). Investor attention and the response of US stock market sectors to the COVID-19 crisis. *Review of Behavioral Finance*, *13*(1), 20–39.

Tetlock, P. C. (2007). Giving content to investor sentiment: The role of media in the stock market. *Journal of Finance*, *62*(3), 1139–1168.

Welch, I., & Goyal, A. (2008). A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies*, *21*(4), 1455–1508.

Zhang, Y.-J., & Wei, Y.-M. (2010). The crude oil market and the gold market: Evidence for cointegration, causality and price discovery. *Resources Policy*, *35*(3), 168–177.